

13-3d Typical Cost Curves

In the examples we have studied so far, the firms have exhibited diminishing marginal product and, therefore, rising marginal cost at all levels of output. This simplifying assumption was useful because it allowed us to focus on the key features of cost curves that are useful in analyzing firm behavior. Yet actual firms are often more complex. In many firms, marginal product does not start to fall immediately after the first worker is hired. Depending on the production process, the second or third worker might have a higher marginal product than the first because a team of workers can divide tasks and work more productively than a single worker. Firms exhibiting this pattern would experience increasing marginal product for a while before diminishing marginal product set in.

Figure 5 shows the cost curves for such a firm, including average total cost (ATC), average fixed cost (AFC), average variable cost (AVC), and marginal cost (MC). At low levels of output, the firm experiences increasing marginal product, and the marginal-cost curve falls. Eventually, the firm starts to experience diminishing marginal product, and the marginal-cost curve starts to rise. This combination of increasing then diminishing marginal product also makes the average-variable-cost curve U-shaped.

Figure 5

Cost Curves for a Typical Firm

Many firms experience increasing marginal product before diminishing marginal product. As a result, they have cost curves shaped like those in this figure. Notice that marginal cost and average variable cost fall for a while before starting to rise.

Despite these differences from our previous example, the cost curves in [Figure 5](#) share the three properties that are most important to remember:

- Marginal cost eventually rises with the quantity of output.
- The average-total-cost curve is U-shaped.
- The marginal-cost curve crosses the average-total-cost curve at the minimum of average total cost.

QuickQuiz

5. A firm is producing **1,000** units at a total cost of **\$5,000**. When it increases production to **1,001** units, its total cost rises to **\$5,008**. For this firm,
- a. marginal cost is **\$5**, and average variable cost is **\$8**.
 - b. marginal cost is **\$8**, and average variable cost is **\$5**.
 - c. marginal cost is **\$5**, and average total cost is **\$8**.
 - d. marginal cost is **\$8**, and average total cost is **\$5**.
6. A firm is producing **20** units with an average total cost of **\$25** and a marginal cost of **\$15**. If it increases production to **21** units, which of the following must occur?

- a. Marginal cost will decrease.
- b. Marginal cost will increase.
- c. Average total cost will decrease.
- d. Average total cost will increase.

7. The government imposes a **\$1,000** per year license fee on all pizza restaurants. As a result, which cost curves shift?

- a. average total cost and marginal cost
- b. average total cost and average fixed cost
- c. average variable cost and marginal cost
- d. average variable cost and average fixed cost

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